Review of "How commutators of constraints reflect the structure of space-time"

C. Teitelboim (1973)

Claudio Pérez.

November 2025

Abstract

In this review, we analyze in detail the classical article by C. Teitelboim, *How commutators of constraints reflect the structure of space-time* (Annals of Physics, 1973). We discuss how the constraints appearing in the reparametrization of a field theory encode the dynamics of the theory itself, and how the Poisson algebra of these constraints codes the causal structure of spacetime.

Contents

1 Introduction **1**

2 Integrability **3**

3 Evaluation of Structure Constants **5**

4 Algebra of Constraints **10**

5 Conclusions **10**

1 Introduction

Claudio Teitelboim's contributions to theoretical physics have proven to be among the most successful and inspiring in Chilean theoretical physics. In the work analized here [1], he analyzes, in a general form, theories on space-like hypersurfaces. The reason for studying this type of hypersurface is that it manifestly reveals the relativistic invariance of a theory, whether in curved or flat spaces [2, 3]. The objective is to set up a general dynamical theory for fields not only on flat Riemannian manifolds but also on curved backgrounds. This defines the fields on a general space-like surface.In the search for a Schrödinger-type quantization on curved surfaces, it is natural to look at the Hamilton-Jacobi equation, where the principal functional S is the classical analog of the quantum wave functional ψ. The observation that, in non-relativistic dynamics, this equation is a first-order PDE led Dirac [4] to the conclusion that we should build a dynamical theory in which canonical variables describing the surface are treated on the same footing as

the dynamical coordinates. In this context, Dirac developed the generalized Hamiltonian formalism.In this formalism, we introduce four extra dynamical variables to the variables describing the fields. In Minkowski spacetime, these surface variables are generally taken to be the four Lorentzian coordinates of a generic point on the surface. However, in general, one could use any curvilinear system following the same procedures, replacing Minkowski with a Riemannian background.

The introduction of these four extra variables leads to four constraints in the Hamiltonian formalism. If the surface variables are $y^A(x)$ (with $A = 0, 1, 2, 3$) and the conjugate momenta are $\pi_A(x)$ ($x = 1, 2, 3$ being curvilinear coordinates on the surface), then the constraints take the form

$$\mathscr{H}_A \equiv \pi_A + K_A = 0. \tag{1}$$

Here K_A is independent of π_A but generally depends on y^A and the canonical variables related to the fields. These four quantities $\mathcal{H}$ contain all the dynamics of the system. Now we consider the change in any functional $\mathbf{F}$ of the canonical variables induced by a deformation of the surface given by $y^A \to y^A + \delta y^A$. This change is determined by the Poisson bracket between the functional and the Hamiltonian

$$\delta H = \int d^3x\, \delta y^A(x) \mathscr{H}_A(x). \tag{2}$$

We can project the constraints into components that are orthogonal ($\mathscr{H}_\perp$) and tangential ($\mathscr{H}_r$) to the surface

$$\mathscr{H}_\perp \equiv \mathscr{H}_A n^A, \tag{3}$$
$$\mathscr{H}_r \equiv \mathscr{H}_A y^A_{,r}, \tag{4}$$

where n^A is the normal vector to the surface. The constraints now adopt the form

$$\mathscr{H}_\perp \equiv \pi_\perp + K_\perp = 0, \tag{5}$$
$$\mathscr{H}_r \equiv \pi_r + K_r = 0, \tag{6}$$

and the Hamiltonian decomposed, with $\delta y^\perp = \epsilon n_A \delta y^A$,$\epsilon = n_B n^B$ and $\delta y^r = g^{rs} \delta y_A y^A_{,s}$

$$\delta H = \int d^3x \left(\delta y^\perp \mathscr{H}_\perp + \delta y^s \mathscr{H}_s \right). \tag{7}$$

This decomposition is common in the context of hypersurfaces and ADM decomposition [5], and the decomposition of the deformations is illustrated in Fig 1.

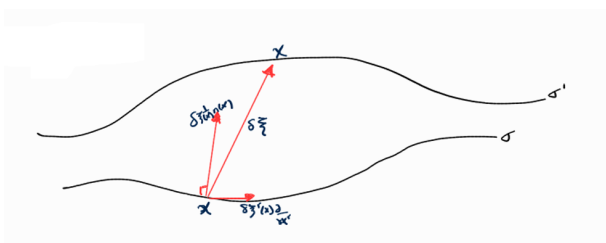

Figure 1: Decomposed deformation vector on the coordinate surface.

The advantage of this decomposition is that a displacement of the surface parallel to itself amounts only to the response of the field variables to a change of coordinates on the surface. This is completely determined by the transformation of the field. We separate the "trivial" part of the problem from the truly dynamical part, which comes from $\mathscr{H}_\perp$ [6]. There is a common aspect to $\mathscr{H}_\perp$ and $\mathscr{H}_r$, even when they vary from one theory to another: the Poisson bracket of any two of them is a linear combination of the constraints themselves. More importantly, this linear combination is the same for all theories. The fact that the bracket of two constraints is a linear combination of the constraints guarantees the preservation of the constraints during the evolution of the system. The commutation rules

$$\{\mathscr{H}_\perp(x), \mathscr{H}_\perp(x')\} = (\mathscr{H}^r(x) + \mathscr{H}^r(x'))\partial_r\delta(x.x'), \tag{8}$$

$$\{\mathscr{H}_r(x), \mathscr{H}_\perp(x')\} = \mathscr{H}_\perp(x)\partial_r\delta(x.x'), \tag{9}$$

$$\{\mathscr{H}_r(x), \mathscr{H}_s(x')\} = (\mathscr{H}_r(x')\partial_s\delta(x.x') + \mathscr{H}_s(x)\partial_r\delta(x.x')) \tag{10}$$

2 Integrability

As stated in the original paper, we assume a field theory (Hamiltonian) with states well-defined on spacelike hypersurfaces embedded in a four-dimensional Riemannian manifold, with coordinates (x^1, x^2, x^3) and a time coordinate, with signature ϵ.Following Dirac's generalized Hamiltonian formalism, for a given functional $\mathbf{F}$ of the canonical variables, the change in $\mathbf{F}$ under a deformation δy^α (with indices corresponding to $\perp$ and spatial components $1, 2, 3$) is given by

$$\delta\mathbf{F} = \int d^3x \left[\mathbf{F}, \delta\mathbf{y}^\alpha\mathcal{H}_\alpha\right], \tag{11}$$

while at the same time the constraint equation holds

$$\mathcal{H}_\alpha = 0. \tag{12}$$

Now we can consider the construction of the commutativity of this deformations. Considering a surface σ and we can go to a infinitesimally shifted (by the deformation) surface σ_1 by the aplication of the deformation $\delta\xi^\alpha$. From this σ_1 we can do the same with another deformation $\delta\eta^\alpha$ and arrive to a surface σ'. If we consider the inverted order in the aplication of the deformations we will arrive to a different surface namely σ''. This situation can bee seen as

$$\begin{aligned} \sigma &\xrightarrow{\delta\xi^\alpha} \sigma_1 \xrightarrow{\delta\eta^\alpha} \sigma', \\ \sigma &\xrightarrow{\delta\eta^\alpha} \sigma_2 \xrightarrow{\delta\xi^\alpha} \sigma''. \end{aligned} \tag{13}$$

The term describing the difference between these two results, arising from alternating the order of application of the deformations, is another deformation, namely $\delta\zeta^\nu$, which is such that it deforms σ' into $\sigma''\sigma''$.

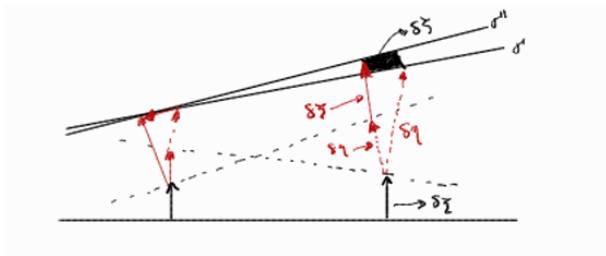

Figure 2: The compensating deformation $\delta\zeta$.

This is called the "compensating deformation"and have the integral form

$$\delta\zeta^{\nu}(x'') = \int d^3x \int d^3x' \kappa^{\nu}_{\alpha\beta}(x''; x, x')\delta\xi^{\alpha}(x)\delta\eta^{\beta}(x') + O((\delta\xi)^2) + O((\delta\eta)^2). \tag{14}$$

An important property is that the exchange of surfaces $\sigma' \leftrightarrow \sigma''$ is equivalent to interchanging $\delta\xi$ with $\delta\eta$. From this, we can see that

$$\kappa^{\nu}_{\alpha\beta}(x''; x, x') = -\kappa^{\nu}_{\beta\alpha}(x''; x', x). \tag{15}$$

We can now determine from Eq. (11) the form that this expression adopts under the two deformations. Let us take the first path ($\sigma \to \sigma_1$) over a functional $\mathbf{F}$ of the canonical variables. The change in $\mathbf{F}$ under the first deformation is, by definition

$$\delta\mathbf{F} = \mathbf{F}[\sigma_1] - \mathbf{F}[\sigma] = \int d^3x\{\mathbf{F}[\sigma], \delta\xi^{\alpha}\mathcal{H}_{\alpha}(x)\}, \tag{16}$$

then

$$\mathbf{F}[\sigma_1] = \mathbf{F}[\sigma] + \int d^3x\{\mathbf{F}[\sigma], \delta\xi^{\alpha}(x)\mathcal{H}_{\alpha}(x)\}, \tag{17}$$

Now if we consider the aplication of the next deformation ($\sigma_1 \to \sigma'$)

$$\begin{aligned}
\mathbf{F}[\sigma'] &= \mathbf{F}[\sigma_1] + \int d^3x'\big\{\mathbf{F}[\sigma_1], \delta\xi^{\beta}(x')\mathcal{H}_{\beta}(x')\big\} \\
&= \mathbf{F}[\sigma] + \int d^3x\{\mathbf{F}, \delta\xi^{\alpha}\mathcal{H}_{\alpha}(x)\} + \int d^3x\{\mathbf{F}, \delta\eta^{\alpha}\mathcal{H}_{\alpha}(x)\} \\
&+ \iint d^3x d^3x'\big\{\{\mathbf{F}, \delta\xi^{\alpha}(x)\mathcal{H}_{\alpha}(x)\}, \delta\eta^{\beta}(x')\mathcal{H}_{\beta}(x')\big\} \\
&= \mathbf{F} + \int d^3x\{\mathbf{F}, \delta\xi^{\alpha}\mathcal{H}_{\alpha} + \delta\eta^{\alpha}\mathcal{H}_{\alpha}\} + \iint d^3x'\big\{\big\{\mathbf{F}, \delta\eta^{\beta}(x')\mathcal{H}_{\beta}\big\}, \delta\xi^{\alpha}(x)\mathcal{H}_{\alpha}(x)\big\}.
\end{aligned} \tag{18}$$

Functionals that do not have explicit dependence are evaluated on the original surface. If we now consider the interchange of the deformations, we obtain the corresponding result by swapping $\delta\xi \leftrightarrow \delta\eta$. Considering the difference between these two functionals and using the Jacobi identity, we find that

$$\begin{aligned}
\mathbf{F}[\sigma''] - \mathbf{F}[\sigma'] = &\int d^3x d^3x'\big\{\{\mathbf{F}, \delta\eta^{\alpha}(x)\mathcal{H}_{\alpha}(x)\}, \delta\xi^{\beta}(x')\mathcal{H}_{\beta}(x')\big\} \\
&- \iint d^3x d^3x'\big\{\{\mathbf{F}, \delta\xi^{\alpha}(x)\mathcal{H}_{\alpha}(x)\}, \delta\eta^{\beta}(x')\mathcal{H}_{\beta}(x')\big\},
\end{aligned} \tag{19}$$

where we consider that

$$\{\{\mathbf{F}, A\}, B\} + \{\{A, B\}, \mathbf{F}\} + \{\{B, \mathbf{F}\}, A\} = 0. \tag{20}$$

So with this consideration our expression reduces to

$$\mathbf{F}[\sigma''] - \mathbf{F}[\sigma'] = \iint d^3x d^3x' \big\{\mathbf{F}, \big\{\delta\eta^\alpha(x)\mathcal{H}_\alpha(x), \delta\xi^\beta(x')\mathcal{H}_\beta(x')\big\}\big\}. \tag{21}$$

This expression corresponds to the variation of the compensating deformation over this functional. This we can also recover from the definition (14) in (11), where we will have

$$\mathbf{F}[\sigma''] - \mathbf{F}[\sigma'] = \int d^3x'' \{\mathbf{F}[\sigma'], \delta\zeta^\nu(x'')\mathcal{H}_\nu(x'')\} \tag{22}$$

$$= \int d^3x'' d^3x' d^3x \big\{\mathbf{F}[\sigma'], \kappa^\nu_{\alpha\beta}(x''; x, x')\delta\xi^\alpha(x)\delta\eta^\beta(x')\mathcal{H}_\nu(x'')\big\}, \tag{23}$$

here we can replace $\mathbf{F}[\sigma']$ from (18) and consider the first order in deformations.

$$\begin{aligned}
&\int d^3x'' d^3x' d^3x \big\{\mathbf{F}[\sigma'], \kappa^\nu_{\alpha\beta}(x''; x, x')\delta\xi^\alpha(x)\delta\eta^\beta(x')\mathcal{H}_\nu(x'')\big\} \\
&= \int d^3x' d^3x \delta\xi^\alpha(x)\delta\eta^\beta(x')\left\{\mathbf{F}[\sigma'], \int d^3x'' \kappa^\nu_{\alpha\beta}(x''; x, x')\mathcal{H}_\nu(x'') + \{\mathcal{H}_\alpha(x'), \mathcal{H}_\beta(x)\}\right\} \\
&\implies \left\{\mathbf{F}, \int d^3x'' \kappa^\nu_{\alpha\beta}(x''; x, x')\mathcal{H}_\nu(x'') - \{\mathcal{H}_\alpha(x'), \mathcal{H}_\beta(x)\}\right\} = 0.
\end{aligned} \tag{24}$$

because the theory will predict a consistent dynamical evolution if and only if (21) and(22) are equal. This is direcly to have a consistent construction from out definition of the compensating deformation. So we can write this requeriment as

$$\left\{\mathbf{F}, \{\mathcal{H}_\alpha(x), \mathcal{H}_\beta(x')\} - \int d^3x'' \kappa^\nu_{\alpha\beta}(x''; x, x')\mathcal{H}_\nu(x'')\right\} = 0. \tag{25}$$

3 Evaluation of Structure Constants

We want to calculate an expression for $\delta\zeta$ in terms of the deformations $\delta\eta$ and $\delta\xi$, and then identify the structure constants $\kappa^\gamma_{\alpha\beta}$. With the fundamental relations for the hypersurface and the full space-time

$$g^{AB} = \epsilon n^A n^B + g^{rs} E^A_r E^B_s, \tag{26}$$

$$n^A n_A = \epsilon, \tag{27}$$

$$n_A E^A_r = 0, \tag{28}$$

where g^{AB} is the metric of the full spacetime with the coordinates $y^A(x)$ and g^{rs} the metric of the hypersurface and $n^A(x)$ the normal vector to it. We can proyect δy^A into a perpendicular and tangential components

$$\delta y^A = \delta y^\perp n^A + \delta y^r E^A_r. \tag{29}$$

Starting form the surface σ defined by $y^A = y^A(x)$ and going to a close surface deformed by $\delta\xi^\alpha$. Then the equation for the new surface is

$$y^A = y^A(x) + \delta\xi^\perp n^A + \delta\xi^r E^A_r, \tag{30}$$

and the normal vectors of the surfaces will differ in a small change $\delta n^A(x)$

$$\delta n^A(x) = \int d^3x' \frac{\delta n^A(x)}{\delta y^B(x')} \left(\delta\xi^\perp(x')n^B(x') + \delta\xi^r(x')E^B_r(x')\right). \tag{31}$$

We can use the fundamental relations in order to see the explicit form of this term. The normal vector to the space like surface is defined by the relations

$$g_{AB}n^A n^B = \epsilon, \tag{32}$$
$$g_{AB}n^A y^B_{,r} = 0. \tag{33}$$

Now we can see that, when we vary w.r.t the coordinates of the spacetime, we will have terms coming from the chain rule and simply factorize this relations by the variations

$$g_{AB,C}\delta y^C n^A n^B + 2n_A\delta n^a = 0, \tag{34}$$
$$g_{AB,C}\delta y^C n^A y^B_{,r} + g_{AB}\left(\delta n^A y^B_{,r} + n^A \delta y^B_{,r}\right) = 0. \tag{35}$$

We can write the decomposition in perpendicular and normal components

$$\delta n^A = \delta n^\perp n^A + \delta n^r y^A_{,r}, \tag{36}$$

with

$$\delta n^\perp = \epsilon n_A \delta n^A, \tag{37}$$
$$\delta n^r = g^{rs} g_{AB}\delta n^A y^B_{,s}, \tag{38}$$

replacing this into (34) and using the properties of the normal vector will drop to zero the last term replacing δn^A. Doing the same in (35) the second term after the replacing will be zero, then we have

$$\delta n^\perp = -\tfrac{1}{2}\,\epsilon\, g_{AB,C}\, n^A n^B\, \delta y^C, \tag{39}$$
$$\delta n^r = -g^{rs}\left(n_A\, \delta y^A{}_{,r} + g_{AB,C}\, n^A y^B{}_{,r}\, \delta y^C\right), \tag{40}$$

and we have the complete form of δn^A in term of the proyections

$$\delta n^D = -g^{rs} n_A y^D_{,s}\delta y^A_{,r} - g_{AB,C} n^A \delta y^C \left(\frac{\epsilon}{2} n^B n^D + g_{rs} y^B_{,r} y^D_{,s}\right). \tag{41}$$

This can be reduced more using decomposed metric and replacing the last term in the parenthesis and renaming the indices

$$g^{AB} = \epsilon n^A n^B + g^{rs} y^A_{,r} y^B_{,s}, \tag{42}$$

then we obtain

$$\delta n^A = -\left[g^{rs} n_B y^A_{,s}\delta y^B_{,r} + g_{CD,B} n^C \delta y^B \left(g^{AD} - \frac{\epsilon}{2} n^A n^D\right)\right]. \tag{43}$$

Then we can see that this expression is factorized by variations of y^A and derivates of it. This expressions looks familiar from variational calculus, so we can think this expression as a variation w.r.t y^A. It is important to remember that, if we vary, the position x can

be changed from one surface to another by the action of the deformations, so we need to add the Dirac delta function to specify that fact in this expression

$$\frac{\delta n^A(x)}{\delta y^B(x')} = -\left[g^{rs} n_B y^A_{,s} \delta_{,r}(x,x') + g_{CD,B} n^C \delta(x,x') \left(g^{AD} - \frac{\epsilon}{2} n^A n^D\right)\right]. \tag{44}$$

Now, we can do the same of (30) to go into another surface by another deformation. From σ we can go to a surface σ' by a deformation $\delta\eta$. We now replace the $y^A(x)$ with the expression (30) and at the same time we replace $\delta\xi^\alpha$ with $\delta\eta^\alpha$ and n^A with $n^A + \delta n^A$

$$\begin{aligned} y^A &= y^A(x) + \delta\xi^\perp n^A + \delta\xi^r y^A_{,r}, \\ y^A &= y^A(x) + \delta\xi^\perp n^A + \delta\xi^r y^A_{,r} + \delta\eta^\perp (n^A + \delta n^A) + \delta\eta^r (\delta\xi^\perp n^A + \delta\xi^s y^A_{,s})_{,r}, \end{aligned} \tag{45}$$

factorizing and using the definition that we obtained for δn^A we have

$$\begin{aligned} y^A = y^A(x) &+ (\delta\xi^\perp + \delta\eta^\perp) n^A + (\delta\xi^r + \delta\eta^r)\, y^A{}_{,r} \\ &+ \delta\eta^\perp \int d^3x' \frac{\delta n^A(x)}{\delta y^B(x')} \left[(\delta\xi^\perp(x') n^B(x') - \delta\xi^r(x')\, y^B{}_{,r}(x'))\right] \\ &+ \delta\eta^r \left(\delta\xi^\perp n^A + \delta\xi^s y^A{}_{,s,r}\right). \end{aligned} \tag{46}$$

If we do the same procedure first deforming with $\delta\eta$ and after with $\delta\xi$, we are going from the initial surface σ to a different final surface σ'', but the expression that we will obtain is the same changing between the deformations only. If we consider deformation that transforms σ' to σ'' with components $\delta\zeta^A$, we can substract (46) with the same equation with $\delta\xi$ and $\delta\eta$ interchanged to know the components of the compensating deformation. Also is better to proyect the components first

$$\delta\zeta^\perp = \epsilon \delta\zeta_A n^A, \tag{47}$$

$$\delta\zeta_r = \delta\zeta_A y^A_{,r}. \tag{48}$$

Considering this with the identities

$$n_c n^c{}_{,r} = -\tfrac{1}{2}\, g_{CD,B}\, n^C n^D\, y^B{}_{,r}, \tag{49}$$

$$n_{B,r}\, y^B{}_{,s} = n_{B,s}\, y^B{}_{,r} = -n_B\, y^B{}_{,sr}. \tag{50}$$

we can calculate $\delta\zeta$ taking the difference between the two paths of deformations $\delta\zeta^A = y^A\big|_{\sigma''} - y^A\big|_{\sigma'}$, wich is equivalent to the difference of (46) with itself with $\delta\xi$ and $\delta\eta$ interchanged. So, taking the difference we start considering the term in the integral and proyecting with the vielbein and using the fundamental relations

$$\frac{\delta n^A(x)}{\delta y^B(x')} e^m_A = -\left[g^{rm} n_B \delta_{,r}(x,x') + g_{CD,B} n^C g^{AD} e^m_A \delta(x,x')\right], \tag{51}$$

so the contribution of the integral will be

$$\begin{aligned} \delta\zeta^m &= \delta\eta^\perp \int d^3x' \frac{\delta n^A(x)}{\delta y^B(x')} \left[(\delta\xi^\perp(x') n^B(x') - \delta\xi^r(x')\, y^B{}_{,r}(x'))\right] - (\delta\xi \longleftrightarrow \delta\eta), \\ &= \delta\eta^\perp \left[-g^{rm} n_B \partial_r \left(\delta\xi^\perp n^B - \delta\xi^r\, y^B{}_{,s}\right) + g_{CD,B} n^C g^{AD} e^m_A \left(\delta\xi^\perp n^B - \delta\xi^r\, y^B{}_{,r}\right)\right] - (\delta\xi \leftrightarrow \delta\eta), \end{aligned}$$

expanding and using fundamental relations we have

$$\delta\eta^\perp\left[-g^{rm}n_Bn^B\partial_r\delta\xi^\perp - g^{rm}(n_B\delta\xi^s_{,r}y^B_{,s} + n_B\delta\xi^s y^B_{,s,r}) - g_{CD,B}n^Cg^{ms}y^B_{,s}\delta\xi^r y^B_{,r}\right] - (\delta\xi \longleftrightarrow \delta\eta)$$
$$= -\epsilon\,\delta\eta^\perp g^{rm}\partial_r\delta\xi^\perp + e^m_A\delta\eta^\perp\delta\xi^r\partial_r n^A - g^{rm}\delta\eta^\perp n_B\delta\xi^s_{,r}y^B_{,s}$$
$$+\epsilon\,\delta\xi^\perp g^{rm}\partial_r\delta\eta^\perp - e^m_A\delta\xi^\perp\delta\eta^r\partial_r n^A + g^{rm}\delta\xi^\perp n_B\delta\eta^s_{,r}y^B_{,s}. \tag{52}$$

Here the terms with $\xi^\perp$ and $\eta^\perp$ cancel out. The term factorized by $g_{CD,B}$ is integrated by parts

$$g_{CD,B}n^Cy^B_{,s} = \partial_B(g_{CD}n^Cy^B_{,s}) - g_{CD}n^C\partial_B(y^B_{,s}), \tag{53}$$

and from the non zero terms from that we can see the cancellation of the third and last term in (52). Taking the difference of (46) (proyected with e^m_A) with itself with the deformations inverted, we have

$$\delta\zeta^m = ... + 0 + ... + \delta\eta^r\left(\delta\xi^\perp n^A_{,r} + \delta\xi^s_{,r}y^A_{,s}\right)e^m_A - \delta\xi^r\left(\delta\eta^\perp n^A_{,r} + \delta\eta^s_{,r}y^A_{,s}\right)e^m_A + int., \tag{54}$$

where "..." terms are terms wich cancell identically when we rest te two terms. The terms "0" are terms that vanish without the difference but by products identities. The "integral" term (fourth term in (46)) is what we obtained proyecting the variation of n^A

$$int = \left[-\epsilon\delta\xi^\perp g^{rm}\partial_r\delta\eta^\perp + \epsilon\delta\eta^\perp g^{rm}\partial_r\delta\xi^\perp\right]. \tag{55}$$

We have left only 4 terms as the proyection of the compensatins deformation

$$\delta\zeta^m = \epsilon\, g^{rs}\big(\delta\eta^\perp\delta\xi^\perp_{,s} - \delta\xi^\perp\delta\eta^\perp_{,s}\big) + \big(\delta\xi^s\delta\eta^\perp_{,s} - \delta\eta^s\delta\xi^\perp_{,s}\big). \tag{56}$$

Next we need to calculate the next proyection of the compensating deformation according to (48).

$$\delta\zeta^\perp = y^A(x)n_A + (\delta\xi^\perp + \delta\eta^\perp)n^An_A + (\delta\xi^r + \delta\eta^r)y^A_{,r}n_A$$
$$+\delta\eta^\perp\int d^3x'\,\frac{\delta n^A(x)}{\delta y^B(x')}\left[(\delta\xi^\perp(x')n^B(x') - \delta\xi^r(x')y^B_{,r}(x'))\right]n_A$$
$$+\delta\eta^r\big(\delta\xi^\perp n^A + \delta\xi^s y^A_{,s,r}\big)n_A - (\delta\xi \longleftrightarrow \delta\eta). \tag{57}$$

We use the fundamental relations to directly see that the first 3 terms will be zero themselves or by the difference under the interchange of deformations. Then we replace the expression (51) into the integral term and obtain

$$\delta\zeta^\perp = \cdots + \cdots + 0 + n_A\delta\eta^\perp\int d^3x'\,\frac{\delta n^A(c)}{\delta y^B(x')}\left(\delta\xi^\perp(x')n^B + \delta\xi^r y^B_{,r}\right) + \delta\eta^r\big(\delta\xi^\perp n^A + \delta\xi^s y^A_{,s,r}\big)n_A - (\delta\xi \leftrightarrow \delta\eta), \tag{58}$$

where we can use again the fundamental relation (49) and obtain

$$\delta\zeta^\perp = \epsilon\delta\eta^r\delta\xi^\perp_{,r} - (\delta\xi \longleftrightarrow \delta\eta), \tag{59}$$

wich we can rewrite as

$$\delta\zeta^\perp = -\left(\delta\eta^r\delta\xi^\perp_{,r} + \delta\xi^r\delta\eta^\perp_{,r}\right). \tag{60}$$

From (14) we want to write each result in the form

$$\delta\zeta^\perp(x'') = \int d^3x\int d^3x'\kappa^\perp_{\alpha\beta}(x'';x,x')\delta\xi^\alpha(x)\delta\eta^\beta(x'), \tag{61}$$

$$\delta\zeta^r(x'') = \int d^3x\int d^3x'\kappa^r_{\alpha\beta}(x'';x,x')\delta\xi^\alpha(x)\delta\eta^\beta(x'). \tag{62}$$

We can use the relation coming from (25). But first, in order to march our expressions iw will be usefull to rewrite the deformations as

$$\delta\xi(x'') = \int d^3x \delta\xi(x)\delta(x'',x), \tag{63}$$

$$\delta\eta(x'') = \int d^3x' \delta\eta(x')\delta(x'',x'). \tag{64}$$

Now we can start with the identification. We consider

$$\begin{aligned}\delta\zeta^\perp &= \delta\xi^r\partial_r\delta\eta^\perp - \delta\eta^r\partial_r\delta\xi^\perp \\ &= \delta\xi^r\int d^3x'\delta\eta^\perp(x')\partial_r\delta(x'',x') - \delta\eta^r\int d^3x\delta\xi^\perp(x)\partial_r\delta(x'',x') \\ &= \int d^3xd^3x'\left[\delta(x'',x)\partial_r\delta(x'',x')\delta\xi^r(x)\delta\eta^\perp(x') - \delta(x'',x')\partial_r\delta(x'',x)\delta\xi^\perp(x)\delta\eta^r(x')\right].\end{aligned} \tag{65}$$

From this relations we can identify directly that from the indices

$$\kappa^\perp_{r\perp}(x'';x,x') = \delta(x'',x)\partial_r\delta(x'',x') = -\kappa^\perp_{\perp r}(x'';x',x). \tag{66}$$

From the first term of (56) we can identify in the same way

$$\begin{aligned}\epsilon g^{rs}\left(\delta\eta^\perp\partial_s\delta\xi^\perp - \delta\xi^\perp\partial_s\delta\eta^\perp\right) &= \epsilon g^{rs}\left[\int d^3x'\delta\eta^\perp(x')\delta(x'',x')\partial_s\int d^3x\delta\xi^\perp(x)\delta(x'',x)\right] \\ = \int d^3x'd^3x\,\epsilon g^{rs}&\left[\delta(x'',x')\partial_s\delta(x'',x)\delta\xi^\perp(x)\delta\eta^\perp(x') - \delta(x'',x)\partial_s\delta(x'',x')\delta\xi^\perp(x)\delta\eta^\perp(x')\right],\end{aligned} \tag{67}$$

From where we can see that

$$\kappa^r_{\perp\perp}(x'';x,x') = \epsilon g^{rs}\left[\delta(x'',x')\partial_s\delta(x'',x) - \delta(x'',x)\partial_s\delta(x'',x')\right]. \tag{68}$$

Finally we do the same procedure with the last term of (56)

$$\begin{aligned}\delta\xi^s\partial_s\delta\eta^r - \delta\eta^s\partial_S\delta\xi^r &= \int d^3x\delta\xi^s(x)\delta(x'',x)\partial_s\int d^3x'\delta\eta^r(x')\delta(x'',x') - (\delta\eta\leftrightarrow\delta\xi) \\ = \int d^3xd^3x'&\left[\delta(x'',x)\partial_s\delta(x'',x')\delta\xi^s(x)\delta^r_p\delta\eta^p(x') - \delta(x'',x')\partial_p\delta(x'',x)\delta^r_s\delta\xi^s(x)\delta\eta^p(x')\right],\end{aligned} \tag{69}$$

from where we identify

$$\kappa^r_{sp}(x'';x,x') = \delta^r_p\delta(x'',x)\partial_s\delta(x'',x') - \delta^r_s\delta(x'',x')\partial_p\delta(x'',x). \tag{70}$$

All the other combinations of structure constants are zero

$$\kappa^r_{s\perp} = 0 \tag{71}$$

$$\kappa^\perp_{\perp\perp} = 0. \tag{72}$$

Resuming the structure constants we have

$$\kappa^{\perp}_{r\perp}(x'';x,x') = -\kappa^{\perp}_{\perp r}(x'';x',x) = \delta(x'',x)\delta_{,r}(x'',x'),$$

$$\begin{aligned}\kappa^{r}_{sp}(x'';x,x') &= -\kappa^{r}_{ps}(x'';x',x)\\ &= \delta(x'',x)\delta_{,s}(x'',x')\delta^r_p - \delta(x'',x')\delta_{,p}(x'',x)\delta^r_s,\end{aligned}$$

$$\begin{aligned}\kappa^{r}_{\perp\perp}(x'';x,x') &= -\kappa^{r}_{\perp\perp}(x'';x',x)\\ &= \epsilon g^{rs}(x'')\left(\delta(x'',x')\delta_{,s}(x'',x) - \delta(x'',x)\delta_{,s}(x'',x')\right),\end{aligned}$$

4 Algebra of Constraints

We can have the algebra of the contraints by simply replacing the structure constant that we find into (25). Replacing first the mixed P.B

$$\begin{aligned}\{\mathscr{H}_r(x),\mathscr{H}_\perp(x')\} &= \int d^3x''\kappa^{\perp}_{r\perp}(x'';x,x')\mathscr{H}_\perp(x'')\\ &= \int d^3x\delta(x'',x)\partial_r\delta(x'',x')\mathscr{H}_\perp(x'') = \mathscr{H}_\perp(x)\partial_r\delta(x,x').\end{aligned} \tag{73}$$

The next one is two perpendicular constraints

$$\begin{aligned}\{\mathscr{H}_\perp,\mathscr{H}_\perp(x')\} &= \int d^3x''\kappa^{r}_{\perp\perp}(x'';x,x')\mathscr{H}_r(x'')\\ &= \int d^3x\epsilon g^{rs}\left[\delta(x'',x')\partial_s\delta(x'',x) - \delta(x'',x)\partial_s\delta(x'',x')\right]\mathscr{H}_r(x''),\end{aligned} \tag{74}$$

and the last P.B is two tangencial constraints

$$\begin{aligned}\{\mathscr{H}_s,\mathscr{H}_p(x')\} &= \int d^3x''\kappa^{r}_{sp}(x'';x,x')\mathscr{H}_r(x'')\\ &= \int d^3x\left[\delta^r_p\delta(x'',x)\partial_s\delta(x'',x') - \delta^r_s\delta(x'',x')\partial_p\delta(x'',x)\right]\mathscr{H}_r(x'')\\ &= \mathscr{H}_p(x)\partial_s\delta(x,x') + \mathscr{H}_s(x')\partial_p\delta(x,x').\end{aligned} \tag{75}$$

In resume of this results we have the algebra of the constraints wich has taken the from

$$\{\mathscr{H}_\perp(x),\mathscr{H}_\perp(x')\} = \varepsilon\left(g^{rs}(x)\,\mathscr{H}_s(x)\,\partial_r\delta(x,x') - g^{rs}(x')\,\mathscr{H}_s(x')\,\partial_r\delta(x',x)\right), \tag{76}$$

$$\{\mathscr{H}_\perp(x),\mathscr{H}_r(x')\} = \mathscr{H}_\perp(x)\,\partial_r\delta(x,x'), \tag{77}$$

$$\{\mathscr{H}_r(x),\mathscr{H}_s(x')\} = \mathscr{H}_s(x)\,\partial_r\delta(x,x') - \mathscr{H}_r(x')\,\partial_s\delta(x',x). \tag{78}$$

This commutations relations are also known from the diffeomorphism algebra but in this case, this is not a true Lie algebra because the coefficients are functions of the phase space.

5 Conclusions

In the original paper and here we have done the same, recover the P.B of the constraints asking for the integrability of Hamilton's equations. An important point is that when

the constraint $\mathscr{H}_r = 0$ is imposed as a restriction on the Hamilton-Jacobi functional, the resulting equation says S does not depend upon the parametrization on the surface. Another important fact is that the constraint algebra have the signature ϵ, so we can read the signature of the hypersurface from this equations also. Maybe the most important result is such that, in order to recover a path independent dynamical evolution, the constraints (25) must hold as C.Teitelboim prove in the original conclusion because the dependence of the structure constants on the field variables, so it is not really a constant. This is what prevents that the deformations form a group.

References

[1] Claudio Teitelboim. How commutators of constraints reflect the space-time structure. *Annals Phys.*, 79:542–557, 1973.

[2] P. Weiss. On the Hamilton-Jacobi theory of mechanics for the generalized variation method. *Proc. Roy. Soc. Lond. A*, 169:102–119, 1938.

[3] P. A. M. Dirac. Quantum theory of localizable dynamical systems. *Phys. Rev.*, 73:1092, 1948.

[4] P. A. M. Dirac. *Lectures on Quantum Mechanics.* Belfer Graduate School of Science, Yeshiva University, New York, 1964.

[5] R. Arnowitt, S. Deser, and C. W. Misner. The Dynamics of General Relativity. In L. Witten, editor, *Gravitation: An Introduction to Current Research*, pages 227–265. Wiley, New York, 1962.

[6] Vincent Moncrief and Claudio Teitelboim. Momentum constraints as integrability conditions for the Hamiltonian constraint. *Phys. Rev. D*, 6:966, 1972.